

1a.Entities: clinics, treatments , procedures, doctors, patients

Relationships: offers, consists, conducted, in charge of, appointments

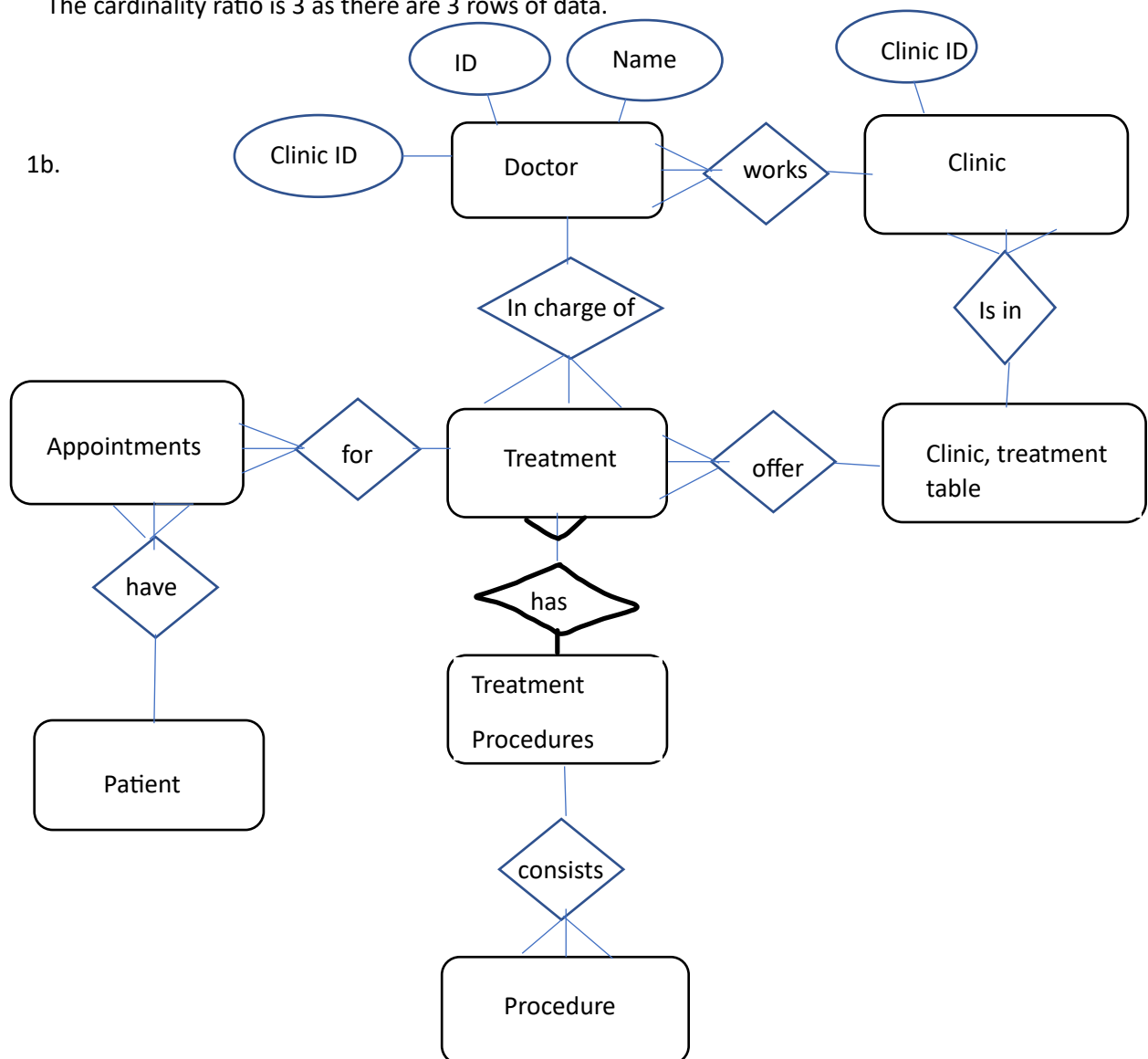
Hospital

Attributes are in the columns

Clinics	treatments	procedures	doctors	patients
1	2	2	DR Hood	Bob
2	1	1	DR James	Sue
3	3	2	DR Sam	Jeff

Schema is {clinic,treatments,procedures,doctors,patients

The cardinality ratio is 3 as there are 3 rows of data.



1c. Student ID: 20453919

```
CREATE TABLE appointment (PatientID int NOT NULL, treatmentID int NOT NULL, DoctorID int NOT NULL, time  
int NOT NULL)
```

```
CREATE TABLE clinic (ClinicID int NOT NULL, treatmentID int NOT NULL, DoctorID int NOT NULL)
```

```
CREATE TABLE doctor (DoctorID int NOT NULL, TreatmentID int NOT NULL, Name varchar(50) NOT NULL, ClinicID  
int NOT NULL)
```

```
CREATE TABLE treatment(treatmentID int NOT NULL,treatmentName varchar(50) NOT NULL, procedureID int  
NOT NULL)
```

```
CREATE TABLE procedure( procedureID int NOT NULL, procedureName varchar(50) NOT NULL)
```

```
CREATE TABLE TreatmentProcedures(procedureID int NOT NULL, treatmentID int NOT NULL)
```

1d.

```
INSERT INTO appointment (PatientID,treatmentID,DoctorID,time)  
VALUES (1001,3,7,1300)
```

1e.

1f.

```
INSERT INTO procedure (procedureID, procedureName)  
VALUES( 17,NewProcedure)
```

Any treatments that use it would then reference it in the proceduretreatments table.

1g.

```
INSERT INTO doctor(DoctorID, TreatmentID , Name, ClinicID )  
Values (34,17,Bob,3)
```

1h.

```
UPDATE doctor, clinic  
SET ClinicID = 2  
WHERE ClinicID = 1
```

1i.

```
DELETE FROM appointment  
WHERE patientID = 3
```

2. HTML,CSS,JAVA

2a.

```
<!DOCTYPE html>
<html>
  <head>
</head>
  <body>
    <ul>
      <li>Text</li>
      <ol>
        <li>Text</li>
        <li>Text</li>
          <ol>
            <li>Text</li>
          </ol>
        </ol>
      </li>
      <li>Text</li>
    </ul>
  </body>
</html>
```

2b.

```
<!DOCTYPE html>
<html>
<head>

<style>
table, th, td {
  border:1px solid black;
}
</style>

</head>
<body>

<table >
  <tr>
    <td> </td>
    <td>00</td>
    <td>00</td>
    <td>00</td>
  </tr>

  <tr>
    <td>00</td>
    <td align = "center" rowspan =2 colspan = 2>00</td>
    <td >00</td>
  </tr>
```

2b continued.

```
<tr>
  <td>00</td>
  <td>00</td>
</tr>
<tr>
  <td>00</td>
  <td>00</td>
  <td>00</td>
  <td></td>
</tr>
</table>
```

```
</body>
</html>
```

2c.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
body {
```

```
  color: blue;
```

```
}
```

```
p {
```

```
  color: red;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
This is some text.
```

```
<p >Text 1</p>
```

```
<p>Text 2</p>
```

```
<p>Text 3</p>
```

```
<p>Text 4</p>
```

```
<p style = color: orange;>Text 5</p>
```

```
This is some more text.
```

```
</body>
```

```
</html>
```

2d.

```
<!DOCTYPE html>
<html>
<head>
<script>

function day(value)
{
if (value = "yes")
{
    body.style = "background-color:Black;";
    p.style = "color:White;";
}
else
{
    body.style = "background-color:White;";
    p.style = "color:Black;";
}
}
}
</script>

</head>
<body style = "background-color:White;">
<p style = color:black> "Hello world"</p>
<form action = day(this)>
<input type="radio" value = "yes">
<input type="submit" value="Submit">
</form>
</body>
</html>
```

2e.

```
<!DOCTYPE html>
<html>
<head>
<script>

function add()
{
    var o = document.getElementById("one").value;
    var t = document.getElementById("two").value;

    var result = Number(o) + Number(t);

    document.getElementById("three").value = result;
}
</script>

</head>
<body style = "background-color:White;">
<p style = color:black> "Hello world"</p>
<form action = day(this)>
<input id = "one" type="text">
<input id = "two" type="text">
<input id = "three" type="text">
<input type="button" onclick = add(>
</form>

</body>
</html>
```

2g.

```
<!DOCTYPE html>
<html>
<script>
function test(btn)
{
var x = Number(btn.value);
btn.value = x+1;
}
</script>
<body>
<input type="button" id="count" value="0" onclick=test(this)>
</body>
</html>
```

2h.

```
<!DOCTYPE html>
<html>
<script>
class person(){
constructor(name) {
    this.name = name;
}
getName(){
    return name;
}
}
function reverse()
{
    if (name1.getName() < name2.getName()){
        swap
    }
}
reverse()

</script>
<body>
<h1>Reverse sorting</h1>
<p class="person">Anna</p>
<p class="person">Chloe</p>
<p class="person">Bob</p>
<p class="person">Darren</p>
</body>
</html>
```

3a.

```
<?php
echo '<!DOCTYPE html>';
echo '<html>';
echo '<head>';
echo '<style>';
echo 'p{';
echo 'color: blue;';
echo '}';
echo '</style>';
echo '</head>';
echo '<body>';
echo '<p>Hello world</p>';
echo '</body>';
echo '</html>';
?>
```

3b.

```
<!DOCTYPE html>
<html>
<body>
```

```
<?php
$x = 15;
echo "<ul>";
while($x >= 6) {
    echo "<li>".$x."</li>";
    $x-=1;
}
echo "</ul>";
?>
```

```
</body>
</html>
```

3c.

```
<?php
$num = $_POST['number'];
if ($num >7)
{
    if ($num <1)
    {
        echo "Not a valid number";
    }
}
else
{
    if ($num = 1)
        echo "Monday";
    }
    if ($num = 2)
        echo "Tuesday";
    }
    if ($num = 3)
        echo "Wednesday";
    }
    if ($num = 4)
        echo "Thursday";
    }
    if ($num = 5)
        echo "Friday";
    }
    if ($num = 6)
        echo "Saturday";
    }
    if ($num = 7)
        echo "Sunday";
    }
?>
```

3d.

```
<?php

    foreach ($_POST as $key => $value) {
        echo $key;
        echo $value;
        echo " ";
    }
?>
```

3d.

```
<?php
```

```
$db_host = 'MySQL';
```

```
    $db_name = 'people';
```

```
    $db_user = 'me';
```

```
    $db_pass = 'password';
```

```
    $conn = new mysqli($db_host, $db_user, $db_pass, $db_name);
```

```
    if ($conn->connect_errno) {
```

```
        die("Failed to connect\n</body>\n</html>");}
```

```
    $sql ="SELECT * FROM Names ;
```

```
    $result = $conn->query($sql);
```

```
<table>
```

```
foreach ($_POST as $key => $value) {
```

```
    echo "<tr>";
```

```
    echo "<td>";
```

```
    echo $key;
```

```
    echo "</td>";
```

```
    echo "<td>";
```

```
    echo $value;
```

```
    echo "</td>";
```

```
</table>
```

```
?>
```